EXPERIMENT 2

EDA – DATA IMPORT AND EXPORT

AIM

To import data from different sources such as CSV files, Excel workbooks, SQL databases and web pages using Python Pandas, inspect the imported data, and export a DataFrame to an Excel file.

SOFTWARE / LIBRARIES REQUIRED

Python 3, Jupyter Notebook or VS Code, Pandas, openpyxl, and (for the optional examples) SQLAlchemy with a suitable database driver, plus Requests and BeautifulSoup for web scraping. Install the libraries that are needed using the following command:

```
pip install pandas openpyxl sqlalchemy requests beautifulsoup4
```

THEORY

Data import is the process of loading data from an external source into a program so that it can be inspected and analysed. Data may be stored in plain-text formats such as CSV, spreadsheet formats such as Excel, relational databases, or web pages. Pandas provides functions such as read_csv() and read_excel() to load tabular data into a DataFrame. A DataFrame stores rows and columns and supports filtering, sorting, summarising and exporting.

Data export is the reverse process: writing the prepared DataFrame to an external file or destination. The to_excel() function writes tabular data to an Excel workbook. The index=False argument prevents Pandas from adding the row index as an extra column. When importing data, the file path, sheet name, encoding, column names and data types should be checked to avoid errors.

ALGORITHM / PROCEDURE

1. Start the Python environment and import the Pandas library.

2. Load a CSV file using read_csv() and an Excel file using read_excel().

3. For an Excel workbook with multiple sheets, specify the required sheet_name.

4. Inspect the loaded DataFrame using head(), shape, columns, info() and isnull().sum().

5. If a database is available, connect to it and read a table or SQL query into a DataFrame.

6. If a permitted public web page contains a suitable HTML table, read the table with read_html() or parse it with a web-scraping library.

7. Export the selected DataFrame to an Excel file using to_excel() with index=False.

8. Open the exported file and verify the rows, columns and data values.

SOURCE CODE

A. Importing data from CSV and Excel

```
import pandas as pd

# Read data from a CSV file
csv_df = pd.read_csv("socialmedia_dataset.csv")
print("CSV preview:")
print(csv_df.head())

# Read data from an Excel workbook
excel_df = pd.read_excel(
    "socialmedia_dataset.xlsx",
    sheet_name=0
)
print("\nExcel preview:")
print(excel_df.head())

# Inspect the DataFrame
print("\nShape:", excel_df.shape)
print("Columns:", list(excel_df.columns))
print("\nData types and non-null counts:")
excel_df.info()
print("\nMissing values:")
print(excel_df.isnull().sum())
```

B. Importing data from a SQL database (example)

Use this example only when a database and its connection details are available. Replace the connection string, table name and credentials with your own settings. Do not publish passwords in a repository.

```
from sqlalchemy import create_engine
import pandas as pd

# Example for a local SQLite database
engine = create_engine("sqlite:///socialmedia.db")

# Load a table into a DataFrame
sql_df = pd.read_sql_query(
    "SELECT * FROM socialmedia",
    engine
)
print(sql_df.head())
```

C. Importing a table from a web page (example)

Use only pages that permit access and reuse. The selected page must contain an HTML table for read_html() to return tabular data.

```
import pandas as pd

url = "https://example.com"
tables = pd.read_html(url)

if tables:
    web_df = tables[0]
    print(web_df.head())
else:
    print("No HTML tables were found.")
```

D. Exporting the DataFrame to Excel

```
# Export the imported Excel data
excel_df.to_excel(
    "socialmedia_export.xlsx",
    index=False
)
print("Data exported successfully.")
```

OUTPUT

The following is an example of the kind of output produced. The actual column names and values depend on the input file used.

```
Excel preview:
   UserID  Age    Gender   Platform
0       1   23    Female   TikTok
1       2   31      Male   Instagram
2       3   28    Female   YouTube
3       4   22      Male   Facebook
4       5   35    Female   TikTok

Shape: (1000, 31)

Data exported successfully.
```

The preview displays the first five rows. The shape gives the number of rows and columns in the selected dataset. The info() method reports column data types and non-null counts, while isnull().sum() reports the number of missing values in each column. After the export command runs, the file socialmedia_export.xlsx is created in the current working directory.

IMPORTANT CHECKS

• Check that the input filename and folder path are correct.

• For Excel files, verify the required worksheet name or index.

• Check data types after import, especially numbers and dates.

• Use index=False when exporting if the row index is not required in Excel.

• For database connections, keep credentials private and use only authorised databases.

• For web data, follow the website's terms and access rules.

RESULT

Data was imported from CSV and Excel files using Pandas. The DataFrame was inspected to view its structure, columns, data types and missing values. Examples were also provided for reading from a SQL database and a web page. Finally, the DataFrame was exported to an Excel file successfully.